

CE 526 : Critical Zone Science

Course Code : CE 526

Course Name : Critical Zone Science

L-P-T-C: 2-1-0-3

Intended for: PG and Ph.D. students

Prerequisites: NA

Mutual Exclusion: NA

Approval : 57th BoA

Course Contents

- **Introduction to critical zone science:** Definition of critical zone and its components, critical zone observatory; Role of soil in critical zone; Introduction to system modelling; Critical zone architecture and evolution 6 Hours
- **Methods in critical zone science:** Design of a critical zone observatory; Tool and techniques for critical zone datasets: Land-atmosphere data, vegetation and associated microbiota, soil (vadose zone), saprolite and bedrock (saturated zone), surface water; Analysis of event-based and continuous fluxes across critical zone interfaces; Pre-processing and analysis of various fluxes using data analysis and statistical computing. 6 Hours
- **Land-atmosphere exchange in critical zone:** Energy budget, earth-atmosphere energy, types of energy transfers, energy balance at earth's surface, effective energy and mass transfer; Carbon budget, carbon cycle and components, global carbon cycle; carbon allocations, controls of Net Primary Production (NPP), carbon relationships 4 hours
- **Water transfer through the critical zone:** Introduction to various runoff generation mechanisms, Dunne diagram; Quantify or track the hydrological cycle in the field; Water balance closure analysis across various temporal and spatial scales; Simulation of runoff generation in hydrologic models; 6 Hours Water balance of a Tree; Water balance impacts, analyze the impact of forest fire and drought on water cycle and nutrient fluxes; Humans in the critical zone; Simulation of water, carbon, energy, and nutrient fluxes 6 Hours

Text books:

1. Giardino, J. R., & Houser, C., Principles and dynamics of the critical zone, Elsevier, 2015.
2. Beven, K. J., Rainfall-runoff modelling: the primer, John Wiley & Sons, 2011.
3. Dingman, S. L., Physical hydrology, Waveland press, 2015.

References:

1. Tarboton, D. G., **Rainfall-runoff processes**, 2003.
2. Banwart, S. A., Nikolaidis, N. P., Zhu, Y. G., Peacock, C. L., & Sparks, D. L., **Soil functions: connecting earth's critical zone**, Annual Review of Earth and Planetary Sciences, 47(1), 333-359, 2019.



3. Relevant research papers will be provided during class.